Projective blowups: a formal and multicentered proof

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Abstract

We give a machine-checked proof, in Lean 4, of the universal property of multicentered blowups: given a finite family of closed subschemes of a scheme $\boldsymbol{X}$, presented on each chart by ideals, there is a scheme $\mathbf{Bl}_{\boldsymbol{Z}}\,\boldsymbol{X}$ over $\boldsymbol{X}$, terminal among $\boldsymbol{X}$-schemes on which every member of the family becomes an effective Cartier divisor. To the best of our knowledge this is the first formalisation of blowups, single- or multicentered, in any theorem prover. The proof is organised around the universal property of the multicentered dilatation of a ring: the initial algebra in which a finite family of ideals becomes generated by non-zero-divisors. Existence and uniqueness of the dilatation are established locally, on affine charts, and every subsequent step of the construction, the base change along open immersions, the comparison isomorphisms on overlaps, the triple overlap maps, the cocycle identity, and the independence of the chosen presentation, is deduced from that single local property by uniqueness alone. We give the exact theorem statement as formalized, the proof architecture, and the main new mathematical ingredients, together with the corresponding Lean code.

Keywords: blowups, dilatations, formalization of mathematics, Lean 4, multi-graded Proj, algebraic geometry, Cartier divisors

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1 Introduction

Blowing up a scheme along a closed subscheme is one of the foundational operations in algebraic geometry. The classical blowup solves a lifting problem: any X-scheme on which the centre becomes a Cartier divisor admits a unique morphism to the blowup.

For a finite family of centres, the multicentered blowup is the universal space on which all the centres become Cartier simultaneously. This paper establishes the universal property of multicentered blowups with a machine-checked proof in Lean 4. To the best of our knowledge this is the first formalisation of blowups, single- or multicentered, in any theorem prover.

The key insight is that the dilatation of a multicenter has a universal property that is simple: it is the initial A-algebra in which each ideal becomes generated by a non-zero-divisor. This property alone drives the entire proof. The global blowup is obtained by gluing dilatation charts using the multigraded Proj construction of Brenner and Schröer, and every step of the globalization follows by uniqueness from that single local property.

The multigraded Proj, the theory of multicentered dilatations of rings, and all supporting machinery had to be developed. The result is a complete, checked proof of a non-trivial theorem in algebraic geometry, with no noetherian, properness, or finite type assumptions on X, and all dependences on the ideals explicit.

This document is organised as follows. Section 1.1 situates the work relative to prior results and existing libraries. Section 1.2 outlines the proof strategy, emphasising the role of the dilatation universal property. Sections 2–4 contain the proof: the theorem statement, the dilatation and its properties, the reduction to the affine case, and the globalisation. Section 5 discusses the formalisation itself and improvements discovered in the proof method.

1.1 Relation to other work

The theorem proved below is not new as a statement, and neither is the dilatation on which its proof rests. The dilatation comes from [6], the published version, where multicentered dilatations of rings, schemes and algebraic spaces are introduced, and where the properties used here as Theorem 13 and Lemmas 11 and 12 are established. The multicentered blowup and its universal property, together with the use of the multigraded Proj of Brenner and Schröer [1] to glue the dilatation charts into a scheme, appeared in the fourth arXiv version of that paper [7], but were dropped from the published version of [6]. A second proof, next to a detailed Brenner–Schröer theory, was given in [8]. For $X = \operatorname{Spec} A$ with closed subschemes $Z_1, \dots, Z_k$ cut out by ideals $L_1, \dots, L_k$, Definition 3.38 sets

$$\operatorname{Bl}_Z X \;:=\; \operatorname{Proj}\bigl(\operatorname{Bl}_{L_1,\dots,L_k} A\bigr),$$

the multi-graded Proj of the multi-Rees algebra, and Proposition 3.39 states that $\operatorname{Bl}_{Z_1,\dots,Z_k} X \to X$ is a final object of the category $\operatorname{Sch}_X^{Z_1,\dots,Z_k\text{-reg}}$ of X-schemes on which all the Z_i become regularly immersed. That is the affine case of Corollary 8, proved below as Theorem 23.

Both ingredients matter, and they enter on an equal footing: the dilatations are what the charts are, and the multi-graded Proj is what makes them into a scheme. For the latter, [7] refers to the multihomogeneous spectra of Brenner and Schröer [1], whose treatment is short and leaves a good deal implicit. That construction was subsequently developed in detail by Mayeux and Riche [8], and it is there that the blowup material from [7] was moved: it does not appear in the published version of [6]. In [8] the statement is proved for general X, as [8, Proposition 3.30]: if $T \to X$ is such that every $T \times_X Y_i$ is an effective Cartier divisor on T, then there is a unique morphism of X-schemes $T \to X'$.

The formalisation of the Brenner–Schröer multi-graded Proj construction that the present work relies on throughout is not our own: it is the one carried out by Mayeux and Zhang in [9] (see also the short account [10]), following the algebraic treatment of [8]. That paper formalizes multi-graded Proj schemes and dilatations of rings in Lean 4, as local, ring-level constructions; it stops short of gluing dilatations of quasi-coherent algebras into a global blowup of schemes, a limitation the authors note explicitly. The present work builds on that formalisation without modification, using it as a black box for everything concerning potions, relevant homogeneous submonoids, and the gluing of Proj from good potion ingredients, and adds the global multicentered blowup and its universal property on top. Mathlib itself contains no theory of blowups, single- or multicentered, so this addition is built entirely from these primitives and from the scheme-theoretic foundations already in mathlib.

The construction below uses both the multi-graded Proj and the dilatations, in the same way as [7]: the dilatations are the affine charts, and the multi-graded Proj of the multi-Rees algebra is what glues them. In the affine case the argument is the one given there, where the charts $D_+(f)$ of $\mathrm{Bl}_Z X$ are identified with the dilatations $A[L_1/f, \dots, L_k/f]$ and the universal property is deduced from the corresponding property of dilatations. Sections 3 and 4.1 to 4.3 are that argument, written out in the form the formalisation uses.

What the present account adds is the globalisation and its verification. The proof of Theorem 6 is organised so that every step beyond the affine case, namely the base change along open immersions, the comparison isomorphisms on overlaps, the triple overlap maps, the cocycle identity, and the independence of the chosen presentation, is deduced from the affine universal property by uniqueness alone, with no further computation in the charts. The bookkeeping of covers and defining ideals that this requires is managed by the notions of presented centre and centre of Section 2.2. This organisation is what makes the argument short enough to formalise, and Section 5 records two places where it produced a better proof than the one we started from.

Multi-graded Proj and blowups of this kind are used elsewhere. Tang [12] constructs Gysin maps for regular immersions in a $\mathbf{P}^1$-unstable, non-$\mathbf{A}^1$-invariant theory of motivic spaces and spectra, using pushout-blowups, and observes there that composing several Gysin maps should require the Proj construction for $\mathbb{N}^n$-graded rings, that is an animated version of the theory of [1] and [8].

The formalisation is built on mathlib [5], which supplies schemes, Spec, fibre products, open and closed immersions, gluing data, homogeneous localisation and the theory of non-zero-divisors. The scheme theory itself goes back to the formalisation of

Buzzard et al. [2], which provided the first example of a non-affine scheme in a theorem prover and on top of which all subsequent scheme-theoretic developments in mathlib, including the present one, are built. Beyond that general infrastructure, none of the material used here was available and all of it had to be developed.

In particular mathlib contains no blowups, in any form. Its file on Rees algebras defines $R[It] = \bigoplus_n I^n t^n$ for a single ideal I, as a subalgebra of $R[X]$, and states in its own header that this "will potentially enable us to calculate some blowup in the future". It contains no dilatations. Its Proj is constructed for $\mathbb{N}$-graded algebras only, so the multigraded Proj that the multicentered theory needs, graded by $\mathbb{Z}^\iota$, is not available either. There is no notion of Cartier divisor, effective or otherwise, and no theory of Weil divisors for schemes. These gaps are the reason for the supporting development described in Section 3.7, and they determine what could be reused and what could not.

Closed subschemes are a partial exception, and since they are what a centre is made of we record the situation precisely. Mathlib defines a closed immersion by the condition that the underlying map is a closed embedding and the induced maps on stalks are surjective. It also knows that this property is local at the target and, on an affine target, equivalent to the source being affine with surjective map on global sections. What it does not develop is the characterisation in terms of ideals chart by chart, which is the form we need, so we take it as our definition of the data attached to a centre. For the reader's convenience we recall the classical statement.

Lemma 1 ([11, Tag 01QN])**.** *Let $i : Z \to X$ be a morphism of schemes. The following are equivalent.*

1. *i is a closed immersion.*
2. *For every affine open $U = \operatorname{Spec} R \subseteq X$ there is an ideal $I \subseteq R$ with $i^{-1}(U) \cong \operatorname{Spec}(R/I)$ as schemes over U.*
3. *There is an affine open cover $X = \bigcup_{j \in J} U_j$, with $U_j = \operatorname{Spec} R_j$, and for each j an ideal $I_j \subseteq R_j$ with $i^{-1}(U_j) \cong \operatorname{Spec}(R_j/I_j)$ as schemes over U_j.*
4. *i induces a homeomorphism of Z onto a closed subset of X and $i^\sharp : \mathcal{O}_X \to i_*\mathcal{O}_Z$ is surjective.*
5. *i induces a homeomorphism of Z onto a closed subset of X, $i^\sharp$ is surjective, and $\ker(i^\sharp) \subseteq \mathcal{O}_X$ is a quasi-coherent sheaf of ideals.*
6. *i induces a homeomorphism of Z onto a closed subset of X, $i^\sharp$ is surjective, and $\ker(i^\sharp) \subseteq \mathcal{O}_X$ is a sheaf of ideals which is locally generated by sections.*

Condition (3) is the one we use. Definition 2 below is that condition with the cover and the ideals promoted from existentially quantified objects to data: rather than asserting that suitable U_j and I_j exist, a presented centre carries a specific choice of them. This is what makes the Cartier condition of Definition 4 expressible, since that condition is a statement about the chosen ideals, and it is why the equivalence relation of Definition 3 has to be imposed afterwards to forget the choice again.

The equivalences of Lemma 1 are not proved in the formalisation and are not needed there. Only the implication from (3) to the existence of the closed subscheme is used, and it is used in the direction that is definitional for us: a presented centre

supplies the data of (3), and the subschemes Z_i are given as part of that data rather than recovered from it.

1.2 Strategy of the proof

The overall method used below is the classical one, and very old: covering, working on charts, and gluing. What is new is the level of detail at which each step is carried out, and the systematic use of one single tool: the universal property of the multicentered dilatation of a ring, to carry out every global step after the reduction to the local affine case.

The proof does not follow [8] closely. The argument there uses multicentered global dilatations of schemes and rests on a body of supporting theory: multi-graded Proj and its charts, quasi-coherent sheaves, closed subschemes, Cartier divisors and their behaviour under base change. Most of that is not available in mathlib, as recorded in Section 1.1. The present proof works at the ring level only, using multicentered dilatations of rings from [6], and constructs the global blowup by hand via covers, charts, gluing data, and uniqueness arguments. Following [8] literally would have meant developing both the global dilatations of schemes and all of their supporting theory. The approach taken here keeps to a minimum what has to be built from scratch. The multi-graded Proj construction of Brenner and Schröer [1] is used to define the blowup by gluing dilatation charts. The universal property of the multicentered dilatation of a ring is then used to prove the universal property of the projective blowup locally, affine case by affine case. Every global step beyond that, including all the gluing and comparison isomorphisms, is governed by the local universal property of the multicentered dilatation alone.

The shape of the argument is the following.

Reduction. A sequence of reduction steps brings the problem down to the affine local situation. One base-changes along an open immersion, restricts to a chart $U_\gamma = \operatorname{Spec} A_\gamma$ of the cover of X carried by the centre, and then restricts further to a chart of the covering supplied by the Cartier hypothesis on T, on which the pulled back ideals are principal and generated by non-zero-divisors. Each of these steps has to be justified, and doing so is a substantial part of the work, because the objects involved are fibre products and the identifications between them are not on the nose.

Existence, locally. In the affine local situation the existence half of the universal property of dilatations applies directly. The hypothesis of that property is, word for word, the Cartier condition: the images of the chosen elements are non-zero-divisors and generate the images of the large ideals. So the required morphism into a chart is produced by Theorem 13, with no further argument.

Gluing, by uniqueness. The local morphisms are then glued, and what makes this work is the uniqueness half of the same universal property. Two local solutions agree on an overlap because both solve the same problem there, so their compatibility is not something to be checked but something to be deduced. The identical mechanism closes every remaining step: the comparison isomorphisms $t_{\gamma\delta}$ on double overlaps, the triple overlap maps $t'_{\gamma\delta\epsilon}$, the cocycle identity, the base change isomorphism of Proposition 24, and the independence of the chosen presentation. In each case two objects are shown to solve one problem, and are therefore canonically identified.

Existence locally, uniqueness globally: that is the whole of the argument. Sections 3 to 4 carry it out.

2 The theorem

2.1 Centres

Throughout, X is a scheme. The object we blow up is not a single closed subscheme but a finite family of them, and part of the data we carry around is a choice of affine chart on which each member of the family is cut out by an honest ideal.

Definition 2. A *presented centre* on X consists of:

1. a type ι of indices;
2. for each $i \in \iota$, a scheme Z_i together with a morphism $Z_i \to X$;
3. an affine open cover $\mathcal{U} = \big(u_\gamma : \operatorname{Spec} A_\gamma \to X\big)_{\gamma \in J}$ of X by open immersions;
4. for each i and γ, an ideal $I_{i\gamma} \subseteq A_\gamma$;
5. for each i and γ, an isomorphism of $\operatorname{Spec} A_\gamma$-schemes

$$\operatorname{Spec}\big(A_\gamma / I_{i\gamma}\big) \xrightarrow{\sim} Z_i \times_X \operatorname{Spec} A_\gamma.$$

Condition (5) does two things at once. It forces $Z_i \times_X U_\gamma$ to be affine, and it says that $I_{i\gamma}$ is the ideal cutting it out. So Z_i is a closed subscheme of X, presented on each chart. In Lean this is a single structure.

`Project/Blowups/PreClosAndClos.lean, lines 52--64`

```
variable (X) in
structure PreClos where
  (indnumb : Type)
  (subscheme: indnumb →Scheme)
  [over : ∀(i : indnumb), Scheme.Over (subscheme i) X]
  cov : Scheme.AffineCover.{u+1} (P := @IsOpenImmersion) X
  ideal: ∀(_ : indnumb) (γ : cov.J), Ideal (cov.obj γ)
  condiso : ∀(i : indnumb) (γ : cov.J),
    Spec (CommRingCat.of (cov.obj γ⧸ideal i γ)) ≅
    pullback (f := subscheme i ↘X) (g := cov.map γ)
  condover : ∀(i : indnumb) (γ : cov.J),
    Scheme.Hom.IsOver (condiso i γ).hom
      (Spec (CommRingCat.of (cov.obj γ)))
```

The cover is auxiliary. Two presented centres should be regarded as the same centre when they cut out the same subschemes, whatever charts they use to do it. This is the relation we quotient by.

Definition 3. Two presented centres Z, Z' are *equivalent* if there is a bijection $\sigma : \iota \to \iota'$ of index types and isomorphisms $Z_i \cong Z'_{\sigma(i)}$ over X for every i. A *centre* on X is an equivalence class. We write $\operatorname{Clos}(X)$ for the set of centres.

`Project/Blowups/PreClosAndClos.lean, lines 73--76, 111, 123--124`

```
structure relStructure (Z Z' : PreClos X) where
  indnumb_equiv : Z.indnumb ≃Z'.indnumb
```

```
  subscheme_iso : ∀i, Z.subscheme i ≅Z'.subscheme (indnumb_equiv i)
  subscheme_iso_over : ∀i, Scheme.Hom.IsOver (subscheme_iso i).hom X

def rel : PreClos X →PreClos X →Prop := fun Z Z' => Nonempty (relStructure Z Z')

variable (X)
def Clos := Quotient (relSetoid X)
```

The covers play no role in the relation. That is the whole point: a centre is a family of closed subschemes of X, and the chart data used to present it is forgotten. Everything we build will be built from a presentation, and a substantial part of the work consists in showing that the result does not depend on the presentation.

2.2 Pullback of centres

Centres pull back along an arbitrary morphism, and the construction is the obvious one, but it is worth writing out because the choice of cover on the target is forced and reappears later.

Let $f : X' \to X$ be a morphism and Z a presented centre on X with cover $(u_\gamma : \operatorname{Spec} A_\gamma \to X)_{\gamma \in J}$. The pulled back centre f^*Z has:

- the same index type ι, with subschemes $Z_i \times_X X'$;
- as cover, the family indexed by pairs (γ, β) where $\gamma \in J$ and β runs over an affine open cover of $X' \times_X U_\gamma$, with chart map

$$\operatorname{Spec} A_{\gamma\beta} \longrightarrow X' \times_X U_\gamma \xrightarrow{\mathrm{pr}_1} X';$$

- as ideal at $(i, (\gamma, \beta))$, the image $I_{i\gamma} \cdot A_{\gamma\beta}$ under the induced ring map $A_\gamma \to A_{\gamma\beta}$.

The point is that one cannot simply pull back the cover: $X' \times_X U_\gamma$ is open in X' but need not be affine, so it has to be covered again. This double indexing is why the index type of a pulled back centre is a sigma type, and it is the reason several later statements quantify over pairs.

That f^* respects the equivalence relation of Definition 3 is straightforward, since the relation only involves the subschemes Z_i and base change of schemes preserves isomorphism over the base. So f^* descends to a map

$$f^* : \operatorname{Clos}(X) \longrightarrow \operatorname{Clos}(X').$$

Functoriality, $(g \circ f)^* = f^* \circ g^*$, holds at the level of Clos though not on the nose at the level of presentations, because the chosen affine covers differ. This is a first instance of a phenomenon that recurs throughout: statements that are false for presentations become true for centres.

`Project/Blowups/PreClosAndClos.lean, lines 351--357, 389--390`

```
def pullback_PreClos (X' : Scheme) (f: X' ⟶X) (Z: PreClos X) : PreClos X' where
  indnumb := Z.indnumb
  subscheme i := pullback (Z.subscheme i ↘X) f
  over i := ⟨pullback.snd _ _⟩
  cov := pull_cov X Z X' f
  ideal i γβ:= pull_ideal X Z X' f γβi
  condiso i γβ:= pullback_PreClos_condiso _ γβi
```

```
def pullback_Clos {X': Scheme} (f: X' ⟶X): Clos X →Clos X' :=
  Quotient.map (pullback_PreClos X X' f) <| fun Z Z' e => Nonempty.map (pullback_lem X Z Z' X'
      f) e
```

2.3 Cartier centres

The condition that drives the whole theory is not invertibility of an ideal sheaf but something slightly more concrete.

Definition 4. A presented centre Z is *Cartier* if for every $i \in \iota$ and every $\gamma \in J$ the ideal $I_{i\gamma} \subseteq A_\gamma$ is principal and generated by a non-zero-divisor. A centre $\mathcal{Z} \in \mathrm{Clos}(X)$ is Cartier if *some* presentation of $\mathcal{Z}$ is Cartier.

`Project/Blowups/PreClosAndClos.lean, lines 134--146`

```
structure PreCars extends PrePri X where
  nonzerodiv : ∀i γ, Submodule.IsPrincipal.generator (ideal i γ) ∈nonZeroDivisors (cov.obj γ)

structure IsPreCars (Z : PreClos X) extends IsPrePri _ Z where
  nonzerodiv : ∀i γ, Submodule.IsPrincipal.generator (Z.ideal i γ) ∈nonZeroDivisors (Z.cov.obj
      γ)

structure IsPri (Z : Clos X) : Prop where
  exists_rep : ∃(Z' : PreClos X), IsPrePri _ Z' ∧Quotient.mk'' Z' = Z

def Pri : Set (Clos X) := {x : Clos X | IsPri _ x}

structure IsCars (Z : Clos X) : Prop where
  exists_rep : ∃(Z' : PreClos X), IsPreCars _ Z' ∧Quotient.mk'' Z' = Z
```

Two features of Definition 4 deserve emphasis, because they are what make the proof work and they are easy to blur in informal writing.

First, the existential quantifier sits outside. A centre is Cartier when there is *one* presentation on which all the ideals are principal on all charts. It is not required that every presentation have this property, and indeed most do not: an arbitrary affine open of a blowup carries an invertible ideal that need not be principal. Getting this quantifier in the right place is what lets us avoid noetherian hypotheses later.

Second, the generator is required to be a non-zero-divisor. Together with principality this says exactly that each Z_i pulls back to an effective Cartier divisor. Requiring only that the ideal be invertible would be weaker in the presence of nilpotents or embedded components, and the dilatation construction of Section 3 produces precisely the non-zero-divisor version, so this is the condition the geometry hands us rather than one we impose for convenience.

2.4 Statement

Fix a presented centre Z on X with ι finite. We will construct a scheme $\mathrm{Bl}_Z X$ over X. The theorem characterising it is the following.

Theorem 5 (Universal property, presented form)**.** *Let Z be a presented centre on X with finite index type. Let T be a scheme over X, with structure morphism $f : T \to X$.*

If the pulled back presented centre f^*Z is Cartier, then there is exactly one morphism

$$\varphi : T \longrightarrow \mathrm{Bl}_Z X$$

over X.

In Lean, with $\mathrm{Bl}_Z X$ written `BlGlob Z`:
`Project/Blowups/BlowupExists.lean, lines 3009--3016`

```
lemma PreProjBlowup_UnivProp
    (Z: PreClos X)
    [DecidableEq Z.indnumb]
    [Fintype Z.indnumb]
    [(i : Z.indnumb →₀ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))]
    {T : Scheme} [T.Over X]
    (cond: IsPreCars _ <| pullback_PreClos _ _ (T ↘X) Z) :
    ∃! φ: T ⟶BlGlob Z, Scheme.Hom.IsOver φX := by
```

The construction depends on the presentation only up to canonical isomorphism, so it descends. Given a centre $\mathcal{Z} \in \mathrm{Clos}(X)$, choose a presentation Z of it and set $\mathrm{Bl}_{\mathcal{Z}} X := \mathrm{Bl}_Z X$.

Theorem 6 (Universal property)**.** *Let $\mathcal{Z} \in \mathrm{Clos}(X)$ be a centre with finite index type and let T be a scheme over X with structure morphism f. If $f^*\mathcal{Z}$ is Cartier, then there is exactly one morphism $T \to \mathrm{Bl}_{\mathcal{Z}} X$ over X.*

`Project/Blowups/BlowupExists.lean, lines 3394--3398`

```
theorem GlobalBlowup_UnivProp (Z : Clos X) [DecidableEq Z.out.indnumb] [Fintype Z.out.indnumb]
    [(i : Z.out.indnumb →₀ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.out.indnumb))]
    {T : Scheme} [T.Over X]
    (cond : IsCars _ <| pullback_Clos (T ↘X) Z) :
    ∃! φ: T ⟶GlobalBlowup Z, Scheme.Hom.IsOver φX := by
```

Alongside these sits the statement that the blowup itself has the property in question.

Theorem 7. *The centre pulled back to $\mathrm{Bl}_Z X$ is Cartier.*

`Project/Blowups/BlowupExists.lean, lines 3342--3345`

```
lemma BlGlob_IsCars (Z : PreClos X)
    [DecidableEq Z.indnumb] [Fintype Z.indnumb]
    [(i : Z.indnumb →₀ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))] :
    IsCars _ (pullback_Clos ((BlGlob Z) ↘X) (Quotient.mk'' Z)) := by
```

The scheme $\mathrm{Bl}_{\mathcal{Z}} X$ itself, for a centre rather than a presentation, is obtained by choosing the representative that the quotient supplies.
`Project/Blowups/BlowupExists.lean, lines 3384--3386`

```
noncomputable def GlobalBlowup (Z : Clos X) [DecidableEq Z.out.indnumb] [Fintype Z.out.indnumb]
    [(i : Z.out.indnumb →₀ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.out.indnumb))] : Scheme :=
  BlGlob Z.out
```

2.5 What the theorem says

Combining Theorems 6 and 7 gives the statement in its cleanest form. Let $\mathcal{C}_{\mathcal{Z}}$ be the category whose objects are X-schemes T such that $f^*\mathcal{Z}$ is Cartier, and whose morphisms are morphisms over X. Then:

Corollary 8. $\mathrm{Bl}_{\mathcal{Z}} X$ *is a terminal object of* $\mathcal{C}_{\mathcal{Z}}$.

So the multicentered blowup is the universal way of making all the Z_i into effective Cartier divisors simultaneously. Three points about this.

It is genuinely simultaneous. One might hope that blowing up several centres could be reduced to blowing up one, either by taking the product ideal $L_1 \cdots L_n$ or by blowing up the L_i one after another, or by forming the fibre product $\mathrm{Bl}_{L_1} X \times_X \cdots \times_X \mathrm{Bl}_{L_n} X$. None of these has the universal property in general. The product ideal remembers too little, iterated blowups depend on the order, and the fibre product is too large. The correct object has charts indexed by the ways of presenting all of the L_i at once, which is what Section 3 produces.

There are no finiteness hypotheses beyond finiteness of ι. No noetherian assumption on X, no coherence assumption on the ideals, no properness or finite type assumption on T. This is possible because the Cartier condition is stated with the existential quantifier over presentations, so the proof never has to promote a locally principal ideal to a principal one on a given chart.

Finally, uniqueness is as important as existence and is used constantly inside the proof. Every comparison isomorphism in the construction, on overlaps, on triple overlaps, and between two presentations of the same centre, is produced by invoking the uniqueness half of an already established universal property rather than by writing down a formula. This is what keeps the gluing under control, and Section 4 is organised around it.

2.6 The classical case

Before going further it is worth checking that Theorem 6 specialises correctly. Take ι a singleton, so the centre is a single closed subscheme $Z \subseteq X$ cut out on U_γ by an ideal I_γ. Then Corollary 8 reads: $\mathrm{Bl}_Z X$ is terminal among X-schemes on which Z becomes an effective Cartier divisor. That is the usual universal property of the blowup of a closed subscheme, as in [3, II.8.1.15] or [11, Section 0806], with two differences worth noting.

The classical statement is usually phrased with the pulled back ideal sheaf being invertible. Ours asks for it to be principal on some affine cover and generated by a non-zero-divisor. For an ideal sheaf on a scheme these agree: an invertible ideal sheaf is locally principal, and its local generators are non-zero-divisors precisely when the corresponding closed subscheme is an effective Cartier divisor. So the two formulations describe the same class of X-schemes.

The second difference is that the classical blowup is usually constructed as Proj of the Rees algebra $\bigoplus_{n\geq 0} I^n$, and the identification with the universal object is a theorem. Here the construction is by dilatation charts, and Section 3.7 shows the two

agree, so nothing is lost. What is gained is that the dilatation description generalises to several centres without modification, whereas the single-graded Rees algebra does not.

3 Dilatation

This section contains the construction that drives everything. The blowup is glued from affine pieces, and each affine piece is a dilatation.

3.1 Multicenters

Definition 9. Let A be a commutative ring. A *multicenter* on A is a triple $F = (\Lambda, (I_j)_{j\in\Lambda}, (a_j)_{j\in\Lambda})$ where Λ is a type, each I_j is an ideal of A and each a_j is an element of A. The *large ideal* at j is

$$\mathcal{L}_j \;:=\; I_j + (a_j).$$

`Project/Dilatation/Multicenter.lean, lines 19--23, 39`

```
@[ext]
structure Multicenter : Type (u+1) where
  (index : Type)
  (ideal : index →Ideal A)
  (elem : index →A)

def LargeIdeal (i : F.index) : Ideal A := F.ideal i + Ideal.span {F.elem i}
```

The pairing of an ideal with a distinguished element is the essential point. We are going to build a ring in which $\mathcal{L}_j$ becomes generated by a_j. Writing $\mathcal{L}_j = I_j + (a_j)$ rather than just taking an ideal and a chosen generator allows a_j to be an arbitrary element of $\mathcal{L}_j$, with I_j absorbing the rest, and this is what lets a single construction cover all the charts.

For $\nu \in \Lambda^{(\mathbb{N})}$, a finitely supported function $\Lambda \to \mathbb{N}$, write

$$\mathcal{L}^\nu := \prod_j \mathcal{L}_j^{\nu_j}, \qquad a^\nu := \prod_j a_j^{\nu_j},$$

both finite products.

3.2 The dilatation algebra

Definition 10. The *dilatation* of A along F is

$$A[F] \;:=\; \left\{ \tfrac{m}{a^\nu} \;:\; \nu \in \Lambda^{(\mathbb{N})},\ m \in \mathcal{L}^\nu \right\} \Big/ \sim,$$

where

$$\frac{m}{a^\nu} \sim \frac{n}{a^\mu} \iff \exists\, \beta \in \Lambda^{(\mathbb{N})} : \quad m\, a^{\beta+\mu} \;=\; n\, a^{\beta+\nu}.$$

Addition and multiplication are the obvious ones,

$$\frac{m}{a^\nu} + \frac{n}{a^\mu} = \frac{ma^\mu + na^\nu}{a^{\nu+\mu}}, \qquad \frac{m}{a^\nu} \cdot \frac{n}{a^\mu} = \frac{mn}{a^{\nu+\mu}},$$

and one checks these are well defined. Note that $ma^\mu \in \mathcal{L}^{\nu+\mu}$ because $a_j \in \mathcal{L}_j$, which is where the definition of the large ideal is used. The map $A \to A[F]$, $x \mapsto x/a^0$, makes $A[F]$ an A-algebra.

`Project/Dilatation/Multicenter.lean, lines 50--56, 89--90`

```
structure PreDil where
  pow : F.index →₀ℕ
  num : A
  num_mem : num ∈F.LargeIdeal ^ pow

def r : F.PreDil →F.PreDil →Prop := fun x y =>
  ∃β: F.index →₀ℕ, x.num * F.elem^(β + y.pow) = y.num * F.elem^(β + x.pow)

variable (F) in
def Dilatation := Quotient F.setoid
```

Informally, $A[F] = A\big[\mathcal{L}_j/a_j : j \in \Lambda\big]$, the subring of the total localisation generated by the fractions ℓ/a_j with $\ell \in \mathcal{L}_j$. The quotient presentation above avoids having to embed anything into a localisation, which matters when the a_j are zero-divisors, and it is the presentation the formalisation uses.

3.3 The two Cartier lemmas

The following pair of statements is the reason dilatations are the right local model. They say that in $A[F]$ the large ideals become principal, generated by non-zero-divisors.

Lemma 11. *For every ν, the image of a^ν in $A[F]$ is a non-zero-divisor.*

Lemma 12. *For every ν, inside $A[F]$ one has*

$$\big(a^\nu\big) \;=\; \mathcal{L}^\nu \cdot A[F].$$

`Project/Dilatation/Multicenter.lean, lines 374, 386--388`

```
lemma nonzerodiv_image (v : F.index →₀ℕ) : algebraMap A A[F] (F.elem^v) ∈nonZeroDivisors A[F]
     := by

lemma image_elem_LargeIdeal_equal (v : F.index →₀ℕ) :
 Ideal.span ({algebraMap A A[F] (F.elem^v)}) =
    Ideal.map (algebraMap A A[F]) (F.LargeIdeal^v):= by
```

The proof of Lemma 11 is short and shows what the relation is doing. Suppose $\frac{m}{a^\mu} \cdot a^\nu = 0$ in $A[F]$. By definition of $\sim$ there is β with $m\,a^\nu a^\beta = 0$ in A. But then taking $\beta' = \nu + \beta$ witnesses $\frac{m}{a^\mu} = 0$. So the element we inverted is a non-zero-divisor not because it was one in A, which it need not be, but because the equivalence relation was set up to kill exactly the elements it annihilates.

Lemma 12 is the inclusion $\mathcal{L}^\nu A[F] \subseteq (a^\nu)$, which holds since $m \in \mathcal{L}^\nu$ gives $m = a^\nu \cdot \frac{m}{a^\nu}$, together with the reverse inclusion, which holds since $a^\nu \in \mathcal{L}^\nu$.

3.4 The universal property

Theorem 13. *Let B be an A-algebra such that*

1. *the image of a_j in B is a non-zero-divisor for every j, and*
2. *$(a_j) = \mathcal{L}_j B$ inside B, for every j.*

Then there is exactly one morphism of A-algebras $A[F] \to B$.

`Project/Dilatation/Multicenter.lean, lines 545--548, 622--625`

```
def desc [Algebra A B]
    (non_zero_divisor : ∀i : F.index, (algebraMap A B) (F.elem i) ∈nonZeroDivisors B)
    (gen : ∀i, Ideal.span {(algebraMap A B) (F.elem i)} = Ideal.map (algebraMap A B)
          (F.LargeIdeal i)) :
    A[F] →ₐ[A] B where

lemma lemma_exists_unique_morphism [Algebra A B]
    (non_zero_divisor : ∀i : F.index, (algebraMap A B) (F.elem i) ∈nonZeroDivisors B)
    (gen : ∀i, Ideal.span {(algebraMap A B) (F.elem i)} = Ideal.map (algebraMap A B)
          (F.LargeIdeal i))
    (χ':A[F]→ₐ[A] B) : χ' = desc F non_zero_divisor gen := by
```

Existence and uniqueness both come from the same observation. Given ν and $m \in \mathcal{L}^\nu$, hypothesis (2) gives $m \in \mathcal{L}^\nu B = (a^\nu)$, so there is $b \in B$ with $m = a^\nu b$, and hypothesis (1) says this b is unique. Send $\frac{m}{a^\nu}$ to that b. Any A-algebra map χ must satisfy $a^\nu \chi\left(\frac{m}{a^\nu}\right) = \chi(m) = m$, so it must send $\frac{m}{a^\nu}$ to the same b. Existence and uniqueness are the two halves of "divide by a^ν, and the quotient is unique".

Now compare Theorem 13 with Definition 4. The hypotheses on B are, word for word, the statement that the centre $(\mathcal{L}_j)_j$ becomes Cartier in B: principal, generated by the image of a_j, which is a non-zero-divisor. So Theorem 13 reads:

> $A[F]$ is the initial A-algebra in which the centre becomes Cartier.

That is the affine case of Corollary 8, in ring-theoretic form. Everything in Section 4 is an elaboration of this one statement: first upgrading it from a single multicenter to a family of ideals with all of its presentations, then from affine to global.

Two special cases are worth recording, both of which follow from Theorem 13 by uniqueness and both of which get used later.

Corollary 14. *If every a_j is already a non-zero-divisor in A and $\mathcal{L}_j = (a_j)$, then $A \to A[F]$ is an isomorphism.*

Indeed $B = A$ then satisfies the hypotheses of Theorem 13, so there is a unique $A[F] \to A$, and both composites with $A \to A[F]$ are identities by the uniqueness clause applied to A and to $A[F]$ in turn.

Corollary 15. *If $F = F'$ as multicenters then $A[F] = A[F']$, canonically and uniquely.*

This looks vacuous and is not, in a formal setting: two multicenters can be equal without the associated types $\mathcal{L}^\nu$ being syntactically identical, and the isomorphism has to be produced. The formalisation does so through the universal property rather than by transport, which avoids dependent type juggling.
`Project/Dilatation/Multicenter.lean, lines 657--658, 673--674`

```
def ofFamilyIso {index : Type} {c : index →A} (c0 : ∀i, c i ∈nonZeroDivisors A) :
    A[ofFamily index c] ≃ₐ[A] A :=

def ofEqual {F F' : Multicenter A} (eq : F = F') :
    A[F] ≃ₐ[A] A[F'] :=
```

3.5 Functoriality

Let $A \to B$ be a ring map and F a multicenter on A. Pushing forward the ideals and the elements gives a multicenter F_B on B with the same index type, ideals $I_j B$ and elements the images of a_j. One checks $\mathcal{L}_j(F_B) = \mathcal{L}_j(F)B$, so the large ideals also push forward.

Proposition 16. *There is a unique morphism of A-algebras $A[F] \to B[F_B]$ compatible with $A \to B$.*

Again this is Theorem 13: in $B[F_B]$ the element a_j is a non-zero-divisor generating $\mathcal{L}_j(F)B[F_B]$, by Lemmas 11 and 12 applied over B. So $A[F] \to B[F_B]$ exists and is unique. The pushed forward multicenter F_B is `image_mult`, and the map is built by `desc`, that is by the universal property.
`Project/Dilatation/Multicenter.lean, lines 772--775, 784--785`

```
def image_mult [Algebra A B] : Multicenter B :=
  { index := _
    ideal i := Ideal.map (algebraMap A B) (F.ideal i)
    elem i := algebraMap A B (F.elem i)}

def functo_dila_alg [Algebra A B]: A[F] →ₐ[A] B[image_mult (B := B) F] :=
  desc F (by
```

When $A \to B$ is flat one can say more: the natural map

$$B \otimes_A A[F] \longrightarrow B[F_B]$$

is an isomorphism. Flatness is genuinely needed here, since the construction of the inverse uses that a non-zero-divisor stays a non-zero-divisor after a flat base change. The repository contains this result, but as explained in Section 4.4 the main line of the proof does not use it, because the comparison it would provide is obtained more cheaply from uniqueness.

3.6 Presentations

We now fix a finite index type ι and a family $L = (L_i)_{i \in \iota}$ of ideals of A. This is the affine centre we want to blow up. A single multicenter is not enough, because a

multicenter comes with a choice of the elements a_j, and there is no canonical choice. The construction takes all choices at once.

Definition 17. A *presentation* of L is a multicenter F on A with finite index type Λ, together with maps $\Psi : \Lambda \to \iota$ and $s : \iota \to \Lambda$ such that

$$\Psi \circ s = \mathrm{id}_\iota, \qquad \mathcal{L}_j = L_{\Psi(j)} \quad \text{for all } j \in \Lambda.$$

We write $\mathcal{M}(L)$ for the collection of presentations.

`Project/Blowups/Bl.lean, lines 18--24`

```
structure Mu : Type (u + 1) where
multicenter : Multicenter A
[fin : Fintype multicenter.index]
Ψ: multicenter.index →ι
sec : ι→multicenter.index
surj : ∀i, Ψ(sec i) = i
cond : ∀i, multicenter.LargeIdeal i = L (Ψ i)
```

Unwinding, a presentation is a finite set Λ of "chart directions", each labelled by the ideal $L_{\Psi(j)}$ it refers to, and for each j a decomposition

$$L_{\Psi(j)} \;=\; I_j + (a_j).$$

The section s requires every L_i to be represented at least once. Nothing requires Ψ to be injective, so an ideal may appear with several different chosen elements.

3.7 Dilatations as charts of a multigraded Proj

The dilatations $A[F]$, $F \in \mathcal{M}(L)$, are the charts of a single scheme. To glue them we realise each as an affine piece of a multigraded Proj.

Let

$$\mathrm{Rees}(L) \;:=\; \bigoplus_{v \in \iota^{(\mathbb{N})}} L^v, \qquad L^v := \prod_i L_i^{v_i},$$

the multi-Rees algebra of the family, graded by $\mathbb{N}^\iota$ and then by $\mathbb{Z}^\iota$ through the inclusion $\mathbb{N}^\iota \hookrightarrow \mathbb{Z}^\iota$. For a homogeneous submonoid S of $\mathrm{Rees}(L)$, let

$$\mathrm{Pot}(S) \;:=\; \Big(S^{-1}\,\mathrm{Rees}(L)\Big)_0$$

be the degree zero part of the homogeneous localisation, called the *potion* of S in the formalisation. The multigraded Proj is glued from the affine schemes $\operatorname{Spec}\mathrm{Pot}(S)$ as S runs over the relevant finitely generated homogeneous submonoids, in the sense of [1].

Given a presentation $F \in \mathcal{M}(L)$, let $S_F \subseteq \mathrm{Rees}(L)$ be the submonoid generated by the elements

$$a_j \in L_{\Psi(j)} = \big(\mathrm{Rees}\, L\big)_{e_{\Psi(j)}}, \qquad j \in \Lambda,$$

that is, a_j placed in degree $e_{\Psi(j)}$, the $\Psi(j)$-th basis vector.

Proposition 18. *There is a canonical isomorphism of A-algebras*

$$A[F] \xrightarrow{\sim} \operatorname{Pot}(S_F).$$

The content is a bookkeeping identity between two descriptions of the same fractions. A homogeneous element of S_F of degree $\nu \in \mathbb{N}^\iota$ is a product a^λ for $\lambda \in \Lambda^{(\mathbb{N})}$ with $\sum_j \lambda_j e_{\Psi(j)} = \nu$. A degree zero element of the localisation $S_F^{-1}\operatorname{Rees}(L)$ is therefore a fraction

$$\frac{m}{a^\lambda}, \qquad m \in \operatorname{Rees}(L)_\nu = L^\nu,$$

and the defining condition $\mathcal{L}_j = L_{\Psi(j)}$ of a presentation says exactly that

$$L^\nu \;=\; \prod_i L_i^{\nu_i} \;=\; \prod_j \mathcal{L}_j^{\lambda_j} \;=\; \mathcal{L}^\lambda.$$

So the numerators available on the two sides coincide. The equivalence relation also matches: two fractions in a homogeneous localisation agree when they agree after multiplying by some element of S_F, and elements of S_F are exactly the a^β, which is the relation of Definition 10. The map itself is again given by the universal property of the dilatation, and the isomorphism is obtained from injectivity and surjectivity, proved separately.

`Project/Blowups/Bl.lean, lines 114--116, 271--272, 605--607`

```
def clo_mu (P: Mu L) :
    HomogeneousSubmonoid (ReesAlgebra.intGrading L):=
  HomogeneousSubmonoid.closure

def clo_mu_mor (P: Mu L) : A[P.multicenter] →ₐ[A] (clo_mu L P).Potion :=
  Multicenter.desc P.multicenter

def Mu_mor_iso (P: Mu L) :
    A[P.multicenter] ≃ₐ[A] (clo_mu L P).Potion :=
  AlgEquiv.ofBijective (clo_mu_mor ..) <| ⟨clo_mu_mor_inj L P, clo_mu_mor_surj L P⟩
```

The condition $\mathcal{L}_j = L_{\Psi(j)}$ is doing all the work. Had we defined a presentation with $\mathcal{L}_j \subseteq L_{\Psi(j)}$, the numerators on the dilatation side would form a smaller set and the map would fail to be surjective. This is another place where pairing an ideal with a chosen element, rather than choosing a generator of an ideal, is what makes the theory fit together.

For the gluing we also need to know that S_F is an admissible index for the Proj construction, meaning finitely generated and relevant. Finite generation is clear, since Λ is finite and S_F is generated by the a_j. Relevance is the condition that the degrees occurring in S_F generate a large enough subgroup of $\mathbb{Z}^\iota$, and it holds because the section $s : \iota \to \Lambda$ guarantees that every basis direction e_i is hit: the element $a_{s(i)}$ has degree e_i. So once again the section, and not just surjectivity of Ψ, is what makes the construction legitimate.

Definition 19. $\mathrm{Bl}^{\mathrm{aff}}(L) :=$ Proj of $\operatorname{Rees}(L)$ taken over the family $\{S_F : F \in \mathcal{M}(L)\}$, that is, the scheme obtained by gluing the affine charts $\operatorname{Spec}\operatorname{Pot}(S_F) \cong \operatorname{Spec} A[F]$,

$F \in \mathcal{M}(L)$, along the standard Proj gluing data. It carries a structure morphism $\mathrm{Bl}^{\mathrm{aff}}(L) \to \operatorname{Spec} A$ induced by $A \cong \mathrm{Rees}(L)_0$.

`Project/Blowups/Bl.lean, lines 703--706, 721--722, 726--728`

```
def map_index (P: Mu L) :
    GoodPotionIngredient (ReesAlgebra.intGrading L) where
  toHomogeneousSubmonoid := clo_mu L P
  relevant := clo_mu_rel L P

def BlMu : Scheme :=
  Proj (τ := Mu L) (map_index L)

instance BlMuOverSpec : Scheme.Over (BlMu L) (Spec <| CommRingCat.of A) where
  hom := (GoodPotionIngredient.over (𝒯 := map_index L)) ≫
    Spec.map (CommRingCat.ofHom <| ReesAlgebra.degreeZeroIso' L)
```

The field `relevant` is the relevance condition discussed above, and the structure morphism to $\operatorname{Spec} A$ is induced by the isomorphism $A \cong \mathrm{Rees}(L)_0$ recorded as `degreeZeroIso'`.

Two remarks. First, the gluing data is inherited from the ambient multigraded Proj, so there is nothing to check: the overlaps and the cocycle condition come for free from the Proj construction. Second, we do not need to know that $\mathrm{Bl}^{\mathrm{aff}}(L)$ is all of $\operatorname{Proj} \mathrm{Rees}(L)$. It is the union of the dilatation charts, and every statement we prove refers only to those charts. This is a deliberate choice and it removes a comparison that would otherwise have to be proved.

To summarise the picture:

$$\begin{array}{ccccc} A[F] & \xrightarrow{\sim} & \mathrm{Pot}(S_F) & \hookrightarrow & \mathrm{Bl}^{\mathrm{aff}}(L) \\ & & & & \downarrow \\ & & & & \operatorname{Spec} A \end{array}$$

Each presentation gives a dilatation, each dilatation is an affine chart, and the charts glue to the affine blowup.

4 Proof of the theorem

The proof has two halves. First the affine case: $\mathrm{Bl}^{\mathrm{aff}}(L)$ is terminal among affine-based $\operatorname{Spec} A$-schemes on which L becomes Cartier. Then the global case: the local blowups over the charts of X are glued, and the gluing is governed at every stage by the affine uniqueness statement.

4.1 The affine centre is Cartier on $\mathrm{Bl}^{\mathrm{aff}}(L)$

The first thing to prove is that $\mathrm{Bl}^{\mathrm{aff}}(L)$ does what it is supposed to do.

Proposition 20. *The centre L pulled back to $\mathrm{Bl}^{\mathrm{aff}}(L)$ is Cartier.*

The presentation exhibiting this is the obvious one: take as cover the family of dilatation charts $\operatorname{Spec} A[F]$, $F \in \mathcal{M}(L)$, indexed by $\mathcal{M}(L)$ itself, and as ideal on the chart F the image $L_i \cdot A[F]$.

For each i and each F, we must produce a generator of $L_i \cdot A[F]$ that is a non-zero-divisor. The section $s : \iota \to \Lambda$ supplies it. Put $j := s(i)$, so $\Psi(j) = i$ and $\mathcal{L}_j = L_i$. Then Lemmas 11 and 12, applied with $\nu = e_j$, give

$$L_i \cdot A[F] \;=\; \mathcal{L}_j \cdot A[F] \;=\; \big(a_j\big), \qquad a_j \text{ a non-zero-divisor in } A[F].$$

So the generator is the dilatation element $a_{s(i)}$. This is visible in the formal proof, which passes exactly these two lemmas to a criterion that checks Cartier-ness chart by chart.

`Project/Blowups/Cars.lean, lines 64--68`

```
lemma BlMuPreClos_IsPreCars (A : CommRingCat) (L : ι→Ideal A) :
    IsPreCars _ (BlMuPreClos A L) := by
  refine isPreCars_of_generators (BlMuPreClos A L)
      (fun i (P : Mu L) => (Mu_mor_iso L P).toRingEquiv
        (algebraMap A A[P.multicenter] (P.multicenter.elem (P.sec i)))) ?_ ?_
```

The presented centre being exhibited, and the resulting Cartier statement, are the following. Its cover is `BlMuAffineCover`, the family of dilatation charts, and its ideal on the chart P is the image of L_i.

`Project/Blowups/Cars.lean, lines 43--48, 89--91`

```
def BlMuPreClos (A : CommRingCat) (L : ι→Ideal A) : PreClos (BlMu L) where
  indnumb := ι
  subscheme i := pullback ((loc_to_PreClos A L).subscheme i ↘Spec A) (BlMu L ↘Spec A)
  over i := ⟨pullback.snd _ _⟩
  cov := BlMuAffineCover A L
  ideal i P := Ideal.map (algebraMap A ((map_index L P).Potion)) (L i)

lemma blowups_Cars (A : CommRingCat) (L : ι→Ideal A) :
    IsCars _ <|
      pullback_Clos (BlMu L ↘(Spec (CommRingCat.of A))) (loc_to_Clos A L) :=
```

The term `P.multicenter.elem (P.sec i)` is $a_{s(i)}$, and `Mu_mor_iso` is the isomorphism of Proposition 18 transporting it into the potion. Nothing else enters. This is the precise sense in which the Cartier property of the blowup is the Cartier property of the dilatation, chart by chart.

The section s is doing real work here. Without it, a presentation could omit some L_i entirely, its chart would carry no chosen generator for that ideal, and Proposition 20 would fail on that chart. This is why Definition 17 demands a section rather than mere surjectivity of Ψ as a property.

A consequence we use repeatedly: an open subscheme of $\mathrm{Bl}^{\mathrm{aff}}(L)$ still has the property, since open immersions are flat and Cartier-ness is stable under flat base change. In the formalisation this is `open_of_blowup_IsCars` (`Project/Blowups/Cars.lean`, lines 100–107).

4.2 Affine uniqueness

Proposition 21. *Let T be a scheme over* $\operatorname{Spec} A$ *on which L becomes Cartier. Then any two morphisms $T \to \mathrm{Bl}^{\mathrm{aff}}(L)$ over* $\operatorname{Spec} A$ *are equal.*

This is the globalisation over T of the uniqueness half of Theorem 13, and the argument is a covering argument. Being equal is local on T, so we may replace T by the members of a cover. By hypothesis L becomes Cartier on T, which by Definition 4 means there is a presentation of the pulled back centre with all ideals principal on non-zero-divisors on the charts of its own cover. Restricting to such a chart $\operatorname{Spec} B \subseteq T$ we are in the situation of Theorem 13: in B each L_i is generated by a non-zero-divisor.

There is one wrinkle. Two morphisms $\operatorname{Spec} B \to \mathrm{Bl}^{\mathrm{aff}}(L)$ need not land in the same chart of $\mathrm{Bl}^{\mathrm{aff}}(L)$, so we cannot immediately apply Theorem 13 to compare them. The formalisation handles this by covering $\operatorname{Spec} B$ further by the preimages of the charts of $\mathrm{Bl}^{\mathrm{aff}}(L)$ under both maps simultaneously, and comparing on the intersections, where both maps factor through a single dilatation and Theorem 13 applies. This is the point of the auxiliary opens appearing in the proof below, which intersect the preimages of two chart ranges with a chart of the covering supplied by the Cartier hypothesis. Note that the hypothesis is stated at the `IsCars` level, so the covering it provides is not one we chose.

`Project/Blowups/UniqueBlowup.lean, lines 254--261`

```
theorem ProjBlowup_UnivProp_unicity_affine
  (A: CommRingCat.{u+1}) (L : ι→Ideal A) [fin : Fintype ι]
  {T : Scheme} [T.Over (Spec A)]
  (cond : IsCars _ (pullback_Clos (T ↘Spec A) (loc_to_Clos A L)))

  (φ φ' : T ⟶BlMu L)
  (φ_over : Scheme.Hom.IsOver φ(Spec A))
  (φ'_over : Scheme.Hom.IsOver φ' (Spec A)) : φ= φ' := by
```

4.3 Affine existence

Proposition 22. *Let T be a scheme over* $\operatorname{Spec} A$ *on which L becomes Cartier. Then there is a morphism $T \to \mathrm{Bl}^{\mathrm{aff}}(L)$ over* $\operatorname{Spec} A$.

Again we work locally on T and glue, and this time the gluing is legitimate precisely because of Proposition 21: local maps that are unique are automatically compatible on overlaps, so they glue with no cocycle computation.

Locally, then, take an affine $\operatorname{Spec} B \subseteq T$ over $\operatorname{Spec} A$ on which each $L_i B$ is generated by a non-zero-divisor c_i. We must produce a map $\operatorname{Spec} B \to \mathrm{Bl}^{\mathrm{aff}}(L)$, that is, an A-algebra map $A[F] \to B$ for a suitable presentation F. The presentation is read off from the data: take $\Lambda = \iota$, $\Psi = \mathrm{id}$, $s = \mathrm{id}$, and for each i choose the decomposition of L_i determined by the generator. Theorem 13 then gives the map, and it is unique.

The actual formal proof (`ProjBlowup_UnivProp_existence_affine_preclo`, `Project/Blowups/BlowupExists.lean`, line 18 onward) is longer than this sketch, and the reason is instructive. The generator c_i produced by the Cartier hypothesis lives in B, not in A, so the presentation it determines is a presentation of LB over B, not of L over A. What we need is a map into $\mathrm{Bl}^{\mathrm{aff}}(L)$, whose charts are dilatations of

presentations over A. The two are bridged by a chain

$$\operatorname{Spec} B \longrightarrow \operatorname{Spec} B[F_B] \longrightarrow \operatorname{Spec} \operatorname{Pot}(S) \longrightarrow \mathrm{Bl}^{\mathrm{aff}}(L),$$

in which the middle terms are potions of homogeneous submonoids of $\mathrm{Rees}(LB)$ and $\mathrm{Rees}(L)$ respectively, and each arrow is an isomorphism or a chart inclusion. Writing this chain out and checking that every stage is a morphism over $\operatorname{Spec} A$ accounts for most of the length of the proof. The mathematical content is the two lines above.

One detail of the chain is where the choice of $\mathcal{M}(L)$ as index set pays off. The submonoid S appearing in the middle is generated by the images of the c_i, and there is no reason for it to be one of the S_F with F a presentation over A. What saves the argument is that $\mathrm{Bl}^{\mathrm{aff}}(L)$ was built with charts indexed by *all* presentations, so it is enough to find some presentation whose chart receives the map, and the decomposition $L_i = I_i + (a_i)$ can be chosen after the fact to match the generator that the Cartier hypothesis handed us. Had we fixed a finite set of charts in advance, as one does when blowing up an ideal with a chosen finite generating set, this step would fail for a general T.

Putting Propositions 20, 21 and 22 together:
`Project/Blowups/BlowupExists.lean, lines 989--993, 1014--1018`

```
lemma ProjBlowup_UnivProp_existence_affine
    (A: CommRingCat) (L : ι→Ideal A) [fin : Fintype ι]
    {T : Scheme} [T.Over (Spec A)]
    (cond : pullback_Clos (T ↘Spec A) (loc_to_Clos A L) ∈CarsAsSubsetOfClos T) :
    ∃φ: T ⟶BlMu L, Scheme.Hom.IsOver φ(Spec A) := by

lemma ProjBlowup_UnivProp_affine
  (A: CommRingCat) (L : ι→Ideal A) [fin : Fintype ι]
  {T : Scheme} [T.Over (Spec A)]
  (cond : pullback_Clos (T ↘Spec A) (loc_to_Clos A L) ∈CarsAsSubsetOfClos T) :
    ∃! φ: T ⟶BlMu L, Scheme.Hom.IsOver φ(Spec A) := by
```

Theorem 23. $\mathrm{Bl}^{\mathrm{aff}}(L)$ *is terminal among* $\operatorname{Spec} A$*-schemes on which* L *becomes Cartier.*

The formalisation packages this as a structure, so that the affine blowup is an object carrying its own universal property rather than a scheme with three separate theorems attached.
`Project/Blowups/UniqueBlowup.lean, lines 38--47`

```
structure conceptual_blowup (Z : Clos X) where
  scheme : Scheme
  over : Scheme.Over scheme X
  in_cars : pullback_Clos (scheme ↘X) Z ∈CarsAsSubsetOfClos scheme
  φ(T : Scheme) [T.Over X] (in_preCars : pullback_Clos (T ↘X) Z ∈(CarsAsSubsetOfClos T)) :
    T ⟶scheme
  φ_over (T : Scheme) [T.Over X] (in_preCars : pullback_Clos (T ↘X) Z ∈(CarsAsSubsetOfClos
      T)) :
    Scheme.Hom.IsOver (φ T in_preCars) X
  φ_uniq (T : Scheme) [T.Over X] (in_preCars : pullback_Clos (T ↘X) Z ∈(CarsAsSubsetOfClos
      T)) :
    ∀φ' : T ⟶scheme, Scheme.Hom.IsOver φ' X →φ' = φT in_preCars
```

Theorem 23 is then the statement that $\mathrm{Bl}^{\mathrm{aff}}(L)$ carries this structure, for the centre L viewed as an element of $\mathrm{Clos}(\operatorname{Spec} A)$.

`Project/Blowups/BlowupExists.lean, lines 1042--1049`

```
def ProjBlowup_is_conceptual_blowups_affine (A: CommRingCat) (L : ι→Ideal A) [fin : Fintype ι] :
    conceptual_blowup (loc_to_Clos A L) where
  scheme := BlMu L
  over := inferInstance
  in_cars := blowups_Cars A L
  φT _ cond := ProjBlowup_φ A L cond
  φ_over T _ cond := ProjBlowup_φ_over A L cond
  φ_uniq T _ cond φ' hφ' := ProjBlowup_φ_uniq A L cond φ' hφ'
```

4.4 Base change along open immersions

To globalise we need to know how $\mathrm{Bl}^{\mathrm{aff}}$ behaves under restriction to an open. The following is the key lemma, and its proof uses no flatness argument at all.

Proposition 24. *Let $A \to B$ be a ring map with $\operatorname{Spec} B \to \operatorname{Spec} A$ an open immersion, and let $L' := (L_i B)_i$. Then there is exactly one isomorphism*

$$\mathrm{Bl}^{\mathrm{aff}}(L) \times_{\operatorname{Spec} A} \operatorname{Spec} B \;\cong\; \mathrm{Bl}^{\mathrm{aff}}(L')$$

over $\operatorname{Spec} B$.

`Project/Blowups/BlowupExists.lean, lines 1337--1341`

```
lemma base_change_Bl_open [Fintype ι] (A B : CommRingCat) [Algebra A B]
    [IsOpenImmersion (Spec B ↘Spec A)] (L: ι→Ideal A) :
    ∃! (e : pullback (BlMu L ↘Spec A) (Spec B ↘Spec A) ≅

    BlMu (L := fun i : ι=> Ideal.map (algebraMap A B) (L i))), Scheme.Hom.IsOver e.hom (Spec B)
        := by
```

One expects to prove this by base changing the dilatation algebras, which requires flatness. That is not what happens. Instead one observes that both sides are terminal for the same problem, and applies Theorem 23 twice.

Write $P := \mathrm{Bl}^{\mathrm{aff}}(L) \times_{\operatorname{Spec} A} \operatorname{Spec} B$. Three Cartier facts are needed:

- L becomes Cartier on P, since $P \to \operatorname{Spec} A$ factors through $\mathrm{Bl}^{\mathrm{aff}}(L)$ and Cartier-ness is stable under base change;
- L' becomes Cartier on P, because L' is the image of L under $A \to B$ and P lies over $\operatorname{Spec} B$, so the two pulled back centres agree;
- L becomes Cartier on $\mathrm{Bl}^{\mathrm{aff}}(L')$, for the same reason read backwards.

Given these, Theorem 23 for $\mathrm{Bl}^{\mathrm{aff}}(L')$ produces a unique $P \to \mathrm{Bl}^{\mathrm{aff}}(L')$ over $\operatorname{Spec} B$, and Theorem 23 for $\mathrm{Bl}^{\mathrm{aff}}(L)$ produces a unique $\mathrm{Bl}^{\mathrm{aff}}(L') \to P$ over $\operatorname{Spec} B$. Both composites are endomorphisms over $\operatorname{Spec} B$ of a scheme on which the centre is Cartier, hence equal to the identity by uniqueness. So the two maps are mutually inverse.

The point that makes this work is that the identification of the two centres, L pulled back and L' pulled back, is an equality in Clos, not an isomorphism that has to be tracked. This is where the quotient of Definition 3 earns its place: Clos is a set,

its elements can be compared with $=$, and the comparison can be rewritten inside a hypothesis. The relevant statement is that the local centre on a chart is the restriction of the global centre, and it is immediate from the definitions because the equivalence relation does not mention covers.

4.5 The local blowups and their overlaps

Now let Z be a presented centre on X, with cover $\mathcal{U} = (u_\gamma : \operatorname{Spec} A_\gamma \to X)_{\gamma \in J}$ and ideals $I_{i\gamma}$. For each γ set

$$B_\gamma \;:=\; \mathrm{Bl}^{\mathrm{aff}}\big((I_{i\gamma})_i\big),$$

the affine blowup of the chart, a scheme over $\operatorname{Spec} A_\gamma$ and hence over X.

For a pair γ, δ let $U_{\gamma\delta} := U_\gamma \times_X U_\delta$ be the overlap, an open of both charts, and let

$$V_{\gamma\delta} \;:=\; B_\gamma \times_{\operatorname{Spec} A_\gamma} U_{\gamma\delta}$$

be its preimage in B_γ. This is an open subscheme of B_γ, and we write $f_{\gamma\delta} : V_{\gamma\delta} \hookrightarrow B_\gamma$ for the inclusion and $\pi_{\gamma\delta} : V_{\gamma\delta} \to U_{\gamma\delta}$ for the projection.
`Project/Blowups/BlowupExists.lean, lines 1393--1400, 1417--1418, 1437--1441, 1614--1620`

```
def ideal_loc (X: Scheme) (Z: PreClos X) (γ : Z.cov.J) : Z.indnumb →Ideal (Z.cov.obj γ) :=
  fun (i : Z.indnumb) => Z.ideal i γ

def Proj_loc (X: Scheme) (Z: PreClos X) (γ : Z.cov.J)
    [DecidableEq Z.indnumb]
    [Fintype Z.indnumb]
    [(i : Z.indnumb →₀ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))] : Scheme :=
  BlMu (A := Z.cov.obj γ) (ideal_loc X Z γ)

def open_pair (X: Scheme) (Z: PreClos X) (γ δ: Z.cov.J) : Scheme :=
  pullback (Z.cov.map γ) (Z.cov.map δ)

def Proj_loc_pair (X: Scheme) (Z: PreClos X) (γ δ: Z.cov.J)
  [DecidableEq Z.indnumb]
    [Fintype Z.indnumb]
  [(i : Z.indnumb →₀ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))] : Scheme :=
    pullback (pullback.fst (Z.cov.map γ) (Z.cov.map δ))

def Proj_loc_pair_mor (X: Scheme) (Z: PreClos X) (γ δ: Z.cov.J)
    [DecidableEq Z.indnumb]
    [Fintype Z.indnumb]
    [(i : Z.indnumb →₀ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))]:
  Proj_loc_pair X Z γδ⟶open_pair X Z γδ:=
    pullback.fst (pullback.fst (Z.cov.map γ) (Z.cov.map δ))
      (Proj_loc X Z γ↘Spec (Z.cov.obj γ))
```

Here `Proj_loc` is B_γ, `open_pair` is $U_{\gamma\delta}$, `Proj_loc_pair` is $V_{\gamma\delta}$, and `Proj_loc_pair_mor` is $\pi_{\gamma\delta}$. The inclusion $f_{\gamma\delta}$ is the other projection of the same fibre product.

Proposition 25. *There is exactly one isomorphism*

$$t_{\gamma\delta} : V_{\gamma\delta} \xrightarrow{\sim} V_{\delta\gamma}$$

over $U_{\gamma\delta}$.

This is the heart of the globalisation, and it is again an application of affine terminality, now on the overlap. Both $V_{\gamma\delta}$ and $V_{\delta\gamma}$ sit over $U_{\gamma\delta}$. Locally on $U_{\gamma\delta}$, say on an affine open $\operatorname{Spec} C \subseteq U_{\gamma\delta}$, Proposition 24 identifies

$$V_{\gamma\delta} \times_{U_{\gamma\delta}} \operatorname{Spec} C \;\cong\; \mathrm{Bl}^{\mathrm{aff}}\big(I_{i\gamma}C\big), \qquad V_{\delta\gamma} \times_{U_{\gamma\delta}} \operatorname{Spec} C \;\cong\; \mathrm{Bl}^{\mathrm{aff}}\big(I_{i\delta}C\big).$$

The two centres appearing on the right are equal, because both are the restriction to $\operatorname{Spec} C$ of the global centre Z: the composites $\operatorname{Spec} C \to U_\gamma \to X$ and $\operatorname{Spec} C \to U_\delta \to X$ agree, being the two projections out of a fibre product over X. So the two sides are the affine blowup of one and the same centre, and Theorem 23 gives a unique isomorphism between them over $\operatorname{Spec} C$.

Uniqueness at each $\operatorname{Spec} C$ is what allows these local isomorphisms to be glued into $t_{\gamma\delta}$, and it also makes $t_{\gamma\delta}$ itself unique. The statement proved in Lean records both the map and its restriction behaviour.
`Project/Blowups/BlowupExists.lean, lines 1963--1973`

```
def Proj_loc_pair_lemm (X: Scheme) (Z: PreClos X) (γ δ: Z.cov.J)
  [DecidableEq Z.indnumb]
  [Fintype Z.indnumb]
  [(i : Z.indnumb →₀ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))]:
  ∃! (f : (Proj_loc_pair X Z γδ) ⟶(Proj_loc_pair X Z δγ)),

  (∃ (pf : Scheme.Hom.IsOver f (open_pair X Z γδ)),
    ∀(C : CommRingCat) (i : Spec C ⟶open_pair X Z γδ) [IsOpenImmersion i],
      restrictToOpen f i =
      (pullbackSymmetry _ _).hom ≫Proj_loc_pair_open_φ X Z γδC i ≫
      (pullbackSymmetry _ _).hom) := by
```

The morphism $t_{\gamma\delta}$ itself is then extracted from that unique existence statement.
`Project/Blowups/BlowupExists.lean, lines 2259--2264`

```
def Proj_loc_pair_swap (X: Scheme) (Z: PreClos X) (γ δ: Z.cov.J)
  [DecidableEq Z.indnumb]
  [Fintype Z.indnumb]
  [(i : Z.indnumb →₀ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))] :
    (Proj_loc_pair X Z γδ) ⟶(Proj_loc_pair X Z δγ) :=
  Classical.choose (Proj_loc_pair_lemm X Z γδ)
```

4.6 Triple overlaps and the cocycle

To glue the B_γ along the $t_{\gamma\delta}$ we must supply the triple overlap data and verify the cocycle condition. Let

$$P_{ijk} \;:=\; V_{ij} \times_{B_i} V_{ik},$$

the preimage in B_i of $U_i \cap U_j \cap U_k$. The required map is

$$t'_{ijk} : P_{ijk} \longrightarrow P_{jki},$$

compatible with t_{ij}, and the cocycle condition is

$$t'_{ijk} \circ t'_{jki} \circ t'_{kij} = \mathrm{id}.$$

The usual construction of t'_{ijk} is topological: one checks that the image of the triple overlap under t_{ij} lies inside the double overlap V_{jk}, and restricts. Here this is not needed, and the reason is worth spelling out because it is the cleanest illustration of how the universal properties organise the proof.

Recall that V_{jk} was defined as a fibre product, $V_{jk} = B_j \times_{\operatorname{Spec} A_j} U_{jk}$, and that U_{jk} is itself a fibre product $U_j \times_X U_k$. So to give a map $P_{ijk} \to V_{jk}$ it is enough to give three maps

$$P_{ijk} \to B_j, \qquad P_{ijk} \to \operatorname{Spec} A_j, \qquad P_{ijk} \to \operatorname{Spec} A_k,$$

agreeing over $\operatorname{Spec} A_j$ and over X respectively. All three are available. The first is

$$P_{ijk} \xrightarrow{\mathrm{pr}_1} V_{ij} \xrightarrow{t_{ij}} V_{ji} \hookrightarrow B_j,$$

the second is its composite with $B_j \to \operatorname{Spec} A_j$, and the third is

$$P_{ijk} \xrightarrow{\mathrm{pr}_2} V_{ik} \xrightarrow{t_{ik}} V_{ki} \hookrightarrow B_k \to \operatorname{Spec} A_k.$$

The compatibility over X holds because t_{ij} and t_{ik} are morphisms over the respective overlaps, so both composites $P_{ijk} \to X$ agree with the structure morphism. So t'_{ijk} is built by the universal property of fibre products alone, with no reference to underlying topological spaces.

`Project/Blowups/BlowupExists.lean, lines 2433--2446`

```
noncomputable def Proj_loc_pair_t' (X : Scheme) (Z : PreClos X) (i j k : Z.cov.J)
    [DecidableEq Z.indnumb] [Fintype Z.indnumb]
    [(i : Z.indnumb →₀ ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))] :
    pullback (Proj_loc_pair_incl X Z i j) (Proj_loc_pair_incl X Z i k) ⟶
      pullback (Proj_loc_pair_incl X Z j k) (Proj_loc_pair_incl X Z j i) :=
  pullback.lift
    (pullback.lift
      (pullback.lift
        (pullback.fst (Proj_loc_pair_incl X Z i j) (Proj_loc_pair_incl X Z i k) ≫
          Proj_loc_pair_swap X Z i j ≫Proj_loc_pair_incl X Z j i ≫
          (Proj_loc X Z j ↘Spec (Z.cov.obj j)))
        (pullback.snd (Proj_loc_pair_incl X Z i j) (Proj_loc_pair_incl X Z i k) ≫
          Proj_loc_pair_swap X Z i k ≫Proj_loc_pair_incl X Z k i ≫
          (Proj_loc X Z k ↘Spec (Z.cov.obj k)))
```

The three nested `pullback.lift`s are exactly the three maps above, assembled into $V_{jk} = B_j \times_{\operatorname{Spec} A_j} (U_j \times_X U_k)$ from the inside out.

For the cocycle, note that all the inclusions $f_{ij} : V_{ij} \to B_i$ are open immersions, hence monomorphisms, so $P_{ijk} \to B_i$ is a monomorphism too and it suffices to prove

$$\left(t'_{ijk} \circ t'_{jki} \circ t'_{kij}\right) \circ m_{ijk} \;=\; m_{ijk}, \qquad m_{ijk} : P_{ijk} \hookrightarrow B_i.$$

Both sides are morphisms $P_{ijk} \to B_i = \mathrm{Bl}^{\mathrm{aff}}\big((I_{\bullet i})\big)$ from an open subscheme of B_i. By Section 4.1 the centre is Cartier on that open, so Proposition 21 applies: it is enough to check the two morphisms agree after composing with $B_i \to \operatorname{Spec} A_i$. Tracing through

the definitions, that computation telescopes and reduces to the single identity

$$t_{ij} \circ t_{ji} = \mathrm{id}_{V_{ij}},$$

which is proved the same way, by the same uniqueness statement, using that a double swap induces the identity on U_{ij} since it is the composite of the two symmetry isomorphisms of a fibre product.

So the entire triple overlap analysis reduces to one instance of affine uniqueness. No intersections of opens are ever computed.

4.7 Gluing and the global universal property

We can now assemble the gluing datum. Recall what has to be supplied: objects B_γ, opens $V_{\gamma\delta} \hookrightarrow B_\gamma$ with $V_{\gamma\gamma} = B_\gamma$, transition isomorphisms $t_{\gamma\delta} : V_{\gamma\delta} \to V_{\delta\gamma}$ with $t_{\gamma\gamma} = \mathrm{id}$, triple overlap maps $t'_{\gamma\delta\epsilon}$ compatible with the $t_{\gamma\delta}$, and the cocycle identity. We have produced all of these:

- B_γ and $V_{\gamma\delta}$ in Section 4.5;
- $t_{\gamma\delta}$ in Proposition 25;
- $t'_{\gamma\delta\epsilon}$ and the cocycle in Section 4.6.

The two degenerate conditions are worth a comment. That $f_{\gamma\gamma}$ is an isomorphism holds because $U_{\gamma\gamma} = U_\gamma \times_X U_\gamma$ equals U_γ, since u_γ is a monomorphism. That $t_{\gamma\gamma} = \mathrm{id}$ then follows from the uniqueness clause of Proposition 25, since the identity is a morphism over $U_{\gamma\gamma}$ and there is only one such. Both are places where an informal account would not pause, and both need the monomorphism property of open immersions to make the two available structure morphisms on $V_{\gamma\gamma}$ agree. Section 5 returns to this.

The data $\left(B_\gamma, V_{\gamma\delta}, f_{\gamma\delta}, t_{\gamma\delta}, t'_{\gamma\delta\epsilon}\right)$ is therefore a gluing datum in the sense of [11], and we define $\mathrm{Bl}_Z X$ to be the resulting scheme, with its morphism to X induced by the $B_\gamma \to \operatorname{Spec} A_\gamma \to X$. The canonical maps $\iota_\gamma : B_\gamma \to \mathrm{Bl}_Z X$ are open immersions whose images cover, and they satisfy

$$\iota_\gamma \circ f_{\gamma\delta} \;=\; \iota_\delta \circ f_{\delta\gamma} \circ t_{\gamma\delta}$$

on $V_{\gamma\delta}$, which is the identity used in the gluing arguments below. The glued scheme is the following, `PreBlGlob` being the gluing datum whose fields are the constructions above.

`Project/Blowups/BlowupExists.lean, lines 2586--2589`

```
abbrev BlGlob (Z: PreClos X) [DecidableEq Z.indnumb]
  [Fintype Z.indnumb]
  [(i : Z.indnumb →₀ ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))] : Scheme :=
  Scheme.GlueData.glued (PreBlGlob Z)
```

Proposition 26. *The centre pulled back to* $\mathrm{Bl}_Z X$ *is Cartier.*

The presentation is assembled from the affine ones. Its cover is the family of all dilatation charts of all the local blowups,

$$\big\{\operatorname{Spec} A_\gamma[F] \;:\; \gamma \in J,\; F \in \mathcal{M}(I_{\bullet\gamma})\big\},$$

indexed by the pairs (γ, F), each mapping into $\operatorname{Bl}_Z X$ as the composite

$$\operatorname{Spec} A_\gamma[F] \hookrightarrow B_\gamma \xrightarrow{\iota_\gamma} \operatorname{Bl}_Z X,$$

a composite of open immersions and hence an open immersion. These charts cover, since the ι_γ cover $\operatorname{Bl}_Z X$ and the dilatation charts cover each B_γ.

The ideals are the ones supplied by Proposition 20, namely $I_{i\gamma}\cdot A_\gamma[F]$, and they are principal on non-zero-divisors for the same reason as there. What has to be checked in addition is that these ideals really do present the pulled back centre, that is, that

$$\operatorname{Spec}\big(A_\gamma[F]/I_{i\gamma}A_\gamma[F]\big) \;\cong\; Z_i \times_X \operatorname{Spec} A_\gamma[F],$$

compatibly with the structure morphisms. This follows by composing the corresponding isomorphism for the presented centre Z over U_γ with the base change of $Z_i \times_X U_\gamma$ along $\operatorname{Spec} A_\gamma[F] \to \operatorname{Spec} A_\gamma$, using that a fibre product taken in two stages agrees with the one taken in a single stage. In the formalisation this composite is written out explicitly, and it is where most of the length of the construction sits.

The Cartier conditions themselves are the affine ones, unchanged. This gives Theorem 7.

`Project/Blowups/BlowupExists.lean, lines 3294--3300`

```
def BlGlobPreClos (Z : PreClos X)
    [DecidableEq Z.indnumb] [Fintype Z.indnumb]
    [(i : Z.indnumb →₀ ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))] :
    PreClos (BlGlob Z) where
  indnumb := Z.indnumb
  subscheme i := pullback (Z.subscheme i ↘X) ((BlGlob Z) ↘X)
  over i := ⟨pullback.snd _ _⟩
```

The choice of cover here is the whole content. A cover of $\operatorname{Bl}_Z X$ by arbitrary affine opens would not do, since the pulled back centre is invertible but not necessarily principal on such an open. It is because the Cartier condition of Definition 4 quantifies existentially over presentations that we are free to use the cover we built the scheme from, on which principality holds by construction.

For the universal property, let $T \to X$ be such that the centre becomes Cartier on T. Set $T_\gamma := T \times_X U_\gamma$. On T_γ the centre $I_{\bullet\gamma}$ becomes Cartier, by restricting the given presentation, so Theorem 23 gives a unique

$$\varphi_\gamma : T_\gamma \longrightarrow B_\gamma$$

over $\operatorname{Spec} A_\gamma$. Composing with $B_\gamma \to \operatorname{Bl}_Z X$ gives maps $T_\gamma \to \operatorname{Bl}_Z X$ over X, and these glue: on $T_\gamma \times_T T_\delta$ both restrictions are morphisms over the overlap into a blowup of a Cartier centre, so they agree by Proposition 21. Note again that the compatibility

is not computed but deduced from uniqueness. This produces $\varphi : T \to \mathrm{Bl}_Z X$ over X, and gives Theorem 5's existence half.

Uniqueness is the same argument with the maps given rather than constructed. If φ, φ' are two morphisms $T \to \mathrm{Bl}_Z X$ over X, they agree after restriction to each T_γ by Proposition 21, hence agree.

4.8 Independence of the presentation

It remains to see that $\mathrm{Bl}_Z X$ depends only on the centre and not on the presentation, which is what allows Theorem 6 to be stated for $\mathcal{Z} \in \mathrm{Clos}(X)$.

Proposition 27. *If Z' and Z'' are equivalent presented centres, then there is exactly one isomorphism $\mathrm{Bl}_{Z'} X \cong \mathrm{Bl}_{Z''} X$ over X.*

`Project/Blowups/BlowupExists.lean, lines 3354--3363`

```
lemma PreProjBlowup_rel (Z' Z'' : PreClos X) (eq: Quotient.mk' Z' = Quotient.mk' Z'')
    {T : Scheme} [T.Over X]
    [DecidableEq Z'.indnumb]
    [Fintype Z'.indnumb]
    [Fintype Z''.indnumb]
    [DecidableEq Z''.indnumb]
    [(i : Z'.indnumb →₀ ℤ) →Decidable (i ∈Set.range (ρNatToInt Z'.indnumb))]
    [(i : Z''.indnumb →₀ ℤ) →Decidable (i ∈Set.range (ρNatToInt Z''.indnumb))]
    (cond: IsPreCars _ <| pullback_PreClos _ _ (T ↘X) Z') :
  ∃! (e : BlGlob Z' ≅BlGlob Z''), Scheme.Hom.IsOver e.hom X := by
```

The hypotheses `T` and `cond` are vestigial: they are not used in the proof, which goes entirely through the two universal properties.

Once Theorem 7 is available this is formal, and it is the same two-sided argument as Proposition 24. Since Z' and Z'' represent the same element of $\mathrm{Clos}(X)$, the centre that becomes Cartier on $\mathrm{Bl}_{Z'} X$ is literally the same element of Clos as the one that becomes Cartier on $\mathrm{Bl}_{Z''} X$. So Theorem 5 applied to Z'' with $T = \mathrm{Bl}_{Z'} X$ gives a unique map one way, the symmetric application gives a unique map the other way, and both composites are identities by uniqueness.

This is where carrying the quotient Clos around pays off. Had we worked only with presented centres, this step would have required comparing the two covers, for instance by passing to a common refinement, and transporting all the chart data across. Because the equivalence relation forgets covers, the two Cartier statements are equalities of elements of a set and can simply be substituted one for the other.

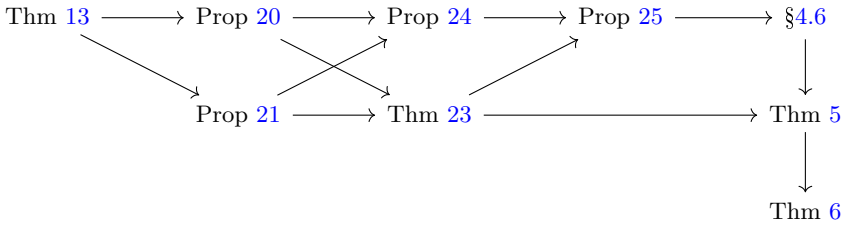
We may therefore define, for a centre $\mathcal{Z} \in \mathrm{Clos}(X)$,

$$\mathrm{Bl}_{\mathcal{Z}} X \;:=\; \mathrm{Bl}_Z X \quad \text{for any presentation } Z \text{ of } \mathcal{Z},$$

well defined up to unique isomorphism over X, and Theorem 6 follows from Theorem 5 by rewriting the Cartier hypothesis along the equality $[\,Z\,] = \mathcal{Z}$. This completes the proof.

4.9 Dependencies

The logical structure of the proof is the following, read left to right.



Everything originates in the universal property of the dilatation. The affine terminality statement, Theorem 23, is used four separate times downstream: to build the base change isomorphism, to build the overlap comparison, to close the cocycle, and to glue the local maps into the global one. Proposition 21, the uniqueness half, is used more often still, since it is what makes each of those constructions canonical and hence compatible with the others.

Note that the diagram has no cycles and no step that depends on the global object being already constructed. This is not automatic. A common way to organise a gluing argument is to construct the object first and verify its properties afterwards, which works informally because one can speak of the object before knowing that it is well defined. Here every arrow points forward, and the global scheme $\mathrm{Bl}_Z X$ is named only after the gluing datum has been completely verified.

4.10 The shape of the argument

The proof used remarkably little computation:

- one genuine construction, the dilatation $A[F]$, with its universal property (Theorem 13);
- one identification, $A[F] \cong \mathrm{Pot}(S_F)$, letting the dilatations be glued by the multigraded Proj machinery;
- one computation, that $L_i A[F] = (a_{s(i)})$ with $a_{s(i)}$ a non-zero-divisor (Section 4.1);
- and then nothing but repeated appeals to uniqueness.

The comparison isomorphisms on overlaps, the triple overlap maps, the cocycle identity, the base change isomorphism, and the independence of the presentation are all instances of the same pattern: two objects solve the same universal problem, therefore there is a unique map between them, therefore the composites are identities. No diagram is chased by hand.

5 Formalisation

The development is written in Lean 4 [4] on top of mathlib [5]. This section explains the parts of the proof that are harder to formalise than to state, and in each case the mathematics comes first.

5.1 What is proved

Every statement quoted in this paper is proved with no unproved assumptions. The check is the standard one: Lean records, for each theorem, the axioms its proof depends on, and for the main theorem and each intermediate result the answer is

`propext`, `Classical.choice`, `Quot.sound`,

which are the three axioms of Lean's ambient logic. In particular `sorryAx`, the axiom that would appear if any part of a proof had been left open, does not occur. This was verified for the main theorem `GlobalBlowup_UnivProp`, for its presented form `PreProjBlowup_UnivProp` and the existence and uniqueness halves separately, for the independence of the presentation `PreProjBlowup_rel`, for the global Cartier property `BlGlob_IsCars`, for the gluing datum `PreBlGlob`, for the affine statements `ProjBlowup_UnivProp_affine` and `ProjBlowup_UnivProp_unicity_affine`, for `blowups_Cars`, and for the overlap and triple overlap constructions `Proj_loc_pair_lemm` and `Proj_loc_pair_t'`.

The repository does contain occurrences of `sorry`, though the claim above is about the theorem rather than about every line of the repository. They are in three places, none of which is reachable from the main theorem. One is `inter_Po` in `Project/Blowups/Bl.lean`, an identity about intersections of chart ranges in the ambient Proj, which nothing in the development uses. One is in `Project/Proj/Sum.lean`, a file that no other file imports. The remainder are in draft material placed after an `#exit` marker, which Lean does not compile, or in a file without a `.lean` extension, which the build system does not see. The axiom check reported above is the reliable statement, since it inspects the proof term of the theorem itself rather than the files around it.

5.2 Reduction to the affine case

Mathematically one says "the statement is local on X, so we may assume X affine". Formally there is no such move, because the object $\mathrm{Bl}_Z X$ does not exist before the reduction: it is built by gluing, and the gluing is part of the theorem. So the informal reduction has to be replayed as an explicit construction, and the order of the development is forced. The affine theory must be complete, including uniqueness, before any global object can be named, and the global universal property can only be stated once the gluing datum has been checked.

This is why the affine uniqueness statement (Proposition 21) appears so early and is used so often. In an informal proof it is a routine remark. Formally it is the tool that replaces every appeal to locality, and it is invoked at five distinct places: the overlap comparison, the cocycle, the base change lemma, the gluing of the local maps, and the independence of the presentation.

5.3 The Cartier condition as data

Definition 4 has an existential quantifier over presentations, and in Lean an existential over a structure carrying data is not a proposition one can casually unfold. The

formalisation therefore keeps two levels apart: `IsPreCars`, a property of a presented centre, which is a plain conjunction of principality and non-zero-divisor statements; and `IsCars`, a property of a class, which asserts the existence of a presentation with `IsPreCars`.

Almost every lemma has to choose which level to use, and getting this wrong makes statements either unprovable or unusable. The pattern that works is: prove things at the `IsPreCars` level where a specific cover is available, and state things at the `IsCars` level where the result must be applied to a cover one does not control. The universal property is stated at the `IsCars` level, because the scheme T in Theorem 6 is arbitrary and no cover of it is given in advance.

There is a genuine mathematical reason not to strengthen `IsCars` to "principal on every affine chart", and formalisation makes it unavoidable to notice. On an arbitrary affine open of $\mathrm{Bl}_Z X$ the pulled back centre is invertible but need not be principal, so the stronger version of the hypothesis would make Theorem 7 false. Any attempt to check the Cartier condition on a canonical cover, rather than on a chosen one, runs into this.

5.4 Building the dilatation without a home

Informally one defines $A[F]$ as a subring of a localisation of A: invert the a_j and take the subring generated by the fractions ℓ/a_j. This is convenient and, in the generality we need, wrong. If some a_j is a zero-divisor then $A \to A[a_j^{-1}]$ is not injective, and the subring description loses information. If some a_j is nilpotent the localisation is zero and the description is vacuous. Since we want no hypotheses on the ideals, neither case can be excluded.

The construction of Definition 10 avoids this by building $A[F]$ directly as a quotient of a set of formal fractions. The cost is that every ring operation has to be defined on representatives and checked to respect the relation, which is several hundred lines of routine verification, and that the ring axioms have to be proved rather than inherited. The benefit is that Lemma 11 becomes available in full generality, and it is the lemma the whole theory rests on.

This is a case where the formal treatment matches what a careful informal account would have to do anyway. The subring-of-a-localisation description is a convenience that holds when the a_j are non-zero-divisors, which is exactly the case in which the dilatation is not doing anything interesting.

5.5 Localisation and potions

Passing from the dilatation to the Proj chart, as in Proposition 18, means comparing two quotients of fraction sets that are defined in different ways. The dilatation quotients formal fractions m/a^ν by an explicit relation; the potion is the degree zero part of a homogeneous localisation, which mathlib constructs as a subquotient of a localisation of the whole graded ring.

The two agree, but not definitionally, and the maps between them are not the identity on representatives. A large part of the supporting development consists of a small calculus for moving between such descriptions: maps induced by an inclusion of

submonoids, by an equality of submonoids, by a ring map, and by multiplying two submonoids together. Each is an isomorphism or a canonical map, each acts on a fraction by rewriting the data attached to it rather than changing the fraction, and the useful fact is that any composite of them can be computed by tracking what happens to a single representative.

Recognising this pattern is what makes the comparison chains in Section 4.3 tractable. Without it, each chain of four or five such maps has to be unfolded by hand, and the intermediate expressions grow past the point of usefulness. With it, a composite is identified by evaluating both sides on one fraction.

The other thing localisation forces into the open is the role of relevance. The Proj construction glues $\operatorname{Spec}\operatorname{Pot}(S)$ over submonoids S that are finitely generated and relevant, and the relevance condition is what guarantees that the resulting scheme has the right dimension and that the charts overlap correctly. In the classical single-graded case relevance is the familiar condition that one uses $D_+(f)$ for f of positive degree, and it is easy to overlook. In the multigraded case it is a condition on the subgroup of $\mathbb{Z}^\iota$ generated by the degrees occurring in S, and as noted in Section 3.7 it holds for S_F exactly because a presentation carries a section $s : \iota \to \Lambda$. Formalising forces this to be checked, and checking it is what reveals that the section is part of the definition of a presentation rather than a convenience.

5.6 Base change and open immersions

Restricting a blowup to an open of the base is the kind of statement that is usually left to the reader. Proposition 24 shows why it is not entirely routine: the natural proof, base changing the dilatation algebra, needs $A \to B$ flat and needs the base change of a dilatation to be a dilatation, which is a real theorem. The proof actually used avoids this completely by observing that both sides are terminal for the same problem.

The formalisation makes the saving concrete. The flat base change route was developed in the repository and is available, but the final proof of Proposition 24 does not call it. It calls the affine universal property three times and the identification of the two centres twice, and nothing else. The general lesson, which is easy to miss informally, is that when two objects both carry universal properties it is usually cheaper to compare the properties than to compare the constructions.

5.7 Open subschemes are not subsets

Informally, $V_{\gamma\delta}$ is "the part of B_γ lying over $U_\gamma \cap U_\delta$", and one manipulates it as a subset. Formally it is a fibre product, and $U_\gamma \cap U_\delta$ is itself a fibre product $U_\gamma \times_X U_\delta$, which is not symmetric in γ and δ on the nose: $U_\gamma \times_X U_\delta$ and $U_\delta \times_X U_\gamma$ are canonically isomorphic but not equal.

Every statement about overlaps therefore carries a symmetry isomorphism that has to be inserted, cancelled, or transported. The relation

$$t_{\gamma\delta} \circ \iota_{\delta\gamma} = \iota_{\gamma\delta} \circ \sigma^{-1}, \qquad \sigma : U_\gamma \times_X U_\delta \xrightarrow{\sim} U_\delta \times_X U_\gamma,$$

appears throughout the development in one form or another. In the special case $\gamma = \delta$ it produces a subtlety that is invisible informally: the scheme $V_{\gamma\gamma}$ carries two different

structure morphisms to $U_{\gamma\gamma}$, the plain one and the one twisted by σ, and these are equal only because u_γ is a monomorphism, so that $\sigma = \mathrm{id}$ for a self-intersection. Verifying $t_{\gamma\gamma} = \mathrm{id}$, which is one of the gluing axioms and which one would not think to check by hand, requires exactly this observation.

5.8 The cocycle

Section 4.6 gave a construction of t'_{ijk} using only fibre products. This was found because the topological argument is hard to formalise and the categorical one is not. Informally, "the triple overlap is contained in the double overlap" is a statement about points and takes a line. Formally it requires computing the image of a morphism of schemes and comparing it with the range of an open immersion, then invoking the lifting property of open immersions along a range inclusion. That is several hundred lines.

The alternative, spelling out that the target is a fibre product of fibre products and giving the three components, is about thirty lines and needs no topology. The mathematical content of the two arguments is the same, but only one of them is short, and the formal setting is what makes the difference visible.

The reduction of the cocycle identity to $t_{ij} \circ t_{ji} = \mathrm{id}$ has a similar flavour. In the category of schemes one cannot compare two morphisms into a glued object by looking at points, so one uses that $P_{ijk} \to B_i$ is a monomorphism and reduces to an identity of morphisms into B_i. Since B_i is a blowup and P_{ijk} an open on which the centre is Cartier, the affine uniqueness statement finishes it. This is a case where the formal proof is arguably more transparent than the informal one, because it makes explicit which universal property is being used.

5.9 Equality of centres versus isomorphism

The quotient $\mathrm{Clos}(X)$ is a quotient of a type of structures by an equivalence relation valued in isomorphisms, so its elements can be compared with equality. This is used heavily, and it is the reason several arguments are short.

In Proposition 25 the two local blowups are compared by observing that their centres are *equal* as elements of $\mathrm{Clos}(\operatorname{Spec} C)$, which allows one hypothesis to be rewritten into the other. Had the centres only been canonically isomorphic, every such step would have required transporting the whole Cartier hypothesis along the isomorphism, which in a dependently typed setting means transporting the covers, the ideals, and the defining isomorphisms as well.

The price is that the quotient has to be built and its interaction with pullback established once and for all: one shows that the base change of presented centres respects the equivalence relation, so that it descends to Clos. That is a small amount of work, done once, and everything downstream benefits.

5.10 Bookkeeping

A large fraction of the development is neither construction nor proof but bookkeeping about structure morphisms. Every object in sight lies over several bases at once. The scheme $V_{\gamma\delta}$ lies over B_γ, over $U_{\gamma\delta}$, over $\operatorname{Spec} A_\gamma$ and over X, and the statements

being proved refer to different ones. In Lean these are typeclass instances, and two morphisms that a mathematician would write the same way can be different terms that happen to be equal.

The recurring difficulty is that a fibre product $Y \times_S T$ carries generic structure morphisms to Y, T and S, but when one needs it as a scheme over something else, say over an affine chart of S, that morphism has to be supplied explicitly and does not agree syntactically with what the generic instances produce. Statements that are true by definition on paper become statements that are true up to an explicit identification.

None of this is mathematically interesting, and it is the bulk of the work. It is also the part that a careful informal write-up silently does correctly, because a mathematician writing $V_{\gamma\delta} \to X$ has one map in mind and never asks which of the four available composites it is.

5.11 Finiteness and decidability hypotheses

The statements carry three hypotheses on the index type ι that a mathematician would not write: that it is finite, that equality on it is decidable, and that membership of $\iota^{(\mathbb{Z})}$ in the image of $\iota^{(\mathbb{N})}$ is decidable. The first is a genuine mathematical hypothesis, used to know that the products L^ν are finite and that the Rees algebra is graded by a finitely generated group. The other two are artefacts, in the sense that they follow from the first together with the law of excluded middle, and they are present because the construction is carried out with data rather than with propositions.

They are not free, though, and they propagate in a way that shapes the statements. The centre $\mathcal{Z} \in \mathrm{Clos}(X)$ of Theorem 6 is an equivalence class, and its index type is only determined up to bijection, so the finiteness and decidability instances have to be attached to a chosen representative. This is why the theorem is stated with hypotheses on $\mathcal{Z}$.out, the representative selected by the quotient. An alternative would be to bundle finiteness into the definition of a centre, which would make the statement cleaner at the cost of making Clos harder to construct.

5.12 What formalising changed

Two arguments in this paper are shorter than the ones we started with, and in both cases the formal setting is what forced the improvement.

The first is the construction of the triple overlap map in Section 4.6. The argument we began with was the standard one, showing that the triple overlap maps into the double overlap and restricting. Making that precise means computing the set-theoretic image of a morphism of schemes and comparing it with the range of an open immersion, which is possible but long. The observation that the target is a fibre product of fibre products, so that the map can be given by its three components with no reference to points, came out of looking for a way to avoid that computation. It is a better argument mathematically as well, since it works verbatim in any category with fibre products.

The second is Proposition 24. The expected proof base changes the dilatation algebra along $A \to B$, which requires $A \to B$ flat and requires knowing that the base change of a dilatation is a dilatation. Both are true and both are in the repository.

Neither is used. Recognising that both sides of the comparison are terminal objects for the same problem replaces the whole argument with three invocations of the affine universal property. Again this is not merely shorter: it explains why the isomorphism is canonical, which the computational proof leaves implicit.

There is a general pattern here. In a setting where every step must be justified, an argument that compares two universal properties is almost always cheaper than one that compares two constructions, because the former needs only the defining property while the latter needs the details of both sides. Working formally makes the cost difference obvious, and so pushes the proof toward the more conceptual of two available arguments.

5.13 Statistics

The development is about 11,000 lines of Lean spread over 48 files. The blowup theory proper occupies roughly 5,500 lines in five files (`Affine.lean`, `PreClosAndClos.lean`, `Cars.lean`, `UniqueBlowup.lean`, `BlowupExists.lean`), with the remainder providing the dilatation algebra, the multigraded Proj construction, the theory of potions and homogeneous submonoids, and additions to mathlib that are independent of blowups. The single largest file is `BlowupExists.lean`, which carries the entire globalisation argument of Sections 4.4 to 4.8.

The imbalance is informative. The affine theory, which contains all of the mathematical novelty, is a small fraction of the total. The globalisation, which an informal account would dispatch in two pages, is the majority. The reason is the one identified above: gluing is where the implicit identifications of an informal argument have to be made explicit, and there are a great many of them.

5.14 Correspondence

For reference, the statements of this paper and the declarations proving them.

Statement	Lean declaration	File
Def. 2	`PreClos`	`PreClosAndClos.lean`
Def. 3	`rel`, `Clos`	`PreClosAndClos.lean`
Def. 4	`IsPreCars`, `IsCars`	`PreClosAndClos.lean`
§2.2	`pullback_PreClos`, `pullback_Clos`	`PreClosAndClos.lean`
Def. 9	`Multicenter`, `LargeIdeal`	`Multicenter.lean`
Def. 10	`PreDil`, `r`, `Dilatation`	`Multicenter.lean`
Lem. 11	`nonzerodiv_image`	`Multicenter.lean`
Lem. 12	`image_elem_LargeIdeal_equal`	`Multicenter.lean`
Thm. 13	`desc`, `lemma_exists_unique_morphism`	`Multicenter.lean`
Cor. 14	`ofFamilyIso`	`Multicenter.lean`
Cor. 15	`ofEqual`	`Multicenter.lean`
Prop. 16	`functo_dila_alg`	`Multicenter.lean`
Def. 17	`Mu`	`Bl.lean`
Prop. 18	`clo_mu_mor`, `Mu_mor_iso`	`Bl.lean`
Def. 19	`BlMu`	`Bl.lean`
Prop. 20	`BlMuPreClos_IsPreCars`, `blowups_Cars`	`Cars.lean`
Prop. 21	`ProjBlowup_UnivProp_unicity_affine`	`UniqueBlowup.lean`
Prop. 22	`ProjBlowup_UnivProp_existence_affine`	`BlowupExists.lean`
Thm. 23	`ProjBlowup_is_conceptual_blowups_affine`	`BlowupExists.lean`
Prop. 24	`base_change_Bl_open`	`BlowupExists.lean`
§4.5	`Proj_loc`, `Proj_loc_pair`	`BlowupExists.lean`
Prop. 25	`Proj_loc_pair_lemm`	`BlowupExists.lean`
§4.6	`Proj_loc_pair_t'`, `PreBlGlob`	`BlowupExists.lean`
§4.7	`BlGlob`	`BlowupExists.lean`
Prop. 26	`BlGlobPreClos`, `BlGlob_IsCars`	`BlowupExists.lean`
Thm. 5	`PreProjBlowup_UnivProp`	`BlowupExists.lean`
Prop. 27	`PreProjBlowup_rel`	`BlowupExists.lean`
Thm. 6	`GlobalBlowup_UnivProp`	`BlowupExists.lean`

Files in `Project/Blowups/` except `Multicenter.lean`, which is in `Project/Dilatation/`.

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